Next-Day 24-Hour Stock Status Prediction

Repository Architecture Analysis, Empirical SOTA Benchmarks, and Model Performance

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Executive Summary

This report focuses **exclusively on next-day 24-hour hourly stock status prediction models** trained on the **FreshRetailNet-50K** dataset.

We present empirical training and validation benchmarks comparing **Linear vs. Non-Linear DLinear Variations**:

1. **Linear DLinear Baselines**: Standard DLinear achieves **0.7932 F1-Score** (6.7k params) and Hourly Slot-Specific DLinear achieves **0.8058 F1-Score**.
2. **Non-Linear DLinear Innovations**: Introducing GELU activation bottlenecks boosts F1-Score to **0.8083** and Stockout Recall to **91.37%**.
3. **Multi-Kernel Non-Linear DLinear**: Multi-scale non-linear trend decomposition ($k = 3, 5, 7$) reaches **0.8156 F1-Score** and the lowest daily stockout duration error of **3.75 hours MAE**.

0.1 Problem Formulation & Dataset Analysis

0.1.1 Unit of Analysis & Target Definition

The task moves beyond coarse daily aggregation to fine-grained hourly stock status forecasting:

- **Input**: Recent N-day historical sequence ($N \in \{7, 10, 14\}$ days) of SKU-store daily metrics.
- **Target**: 24 binary hourly stock-status values for the subsequent day ($t + 1$).
- **Unit of Analysis**: SKU-store time series created by combining `series_id = city_id + store_id + product_id`.

0.2 Architectural Impact: Linear vs. Non-Linear DLinear Variations

To evaluate the effect of non-linearity on series decomposition models, we benchmarked GELU, SwiGLU, and Multi-Kernel non-linear DLinear variants against standard linear models:

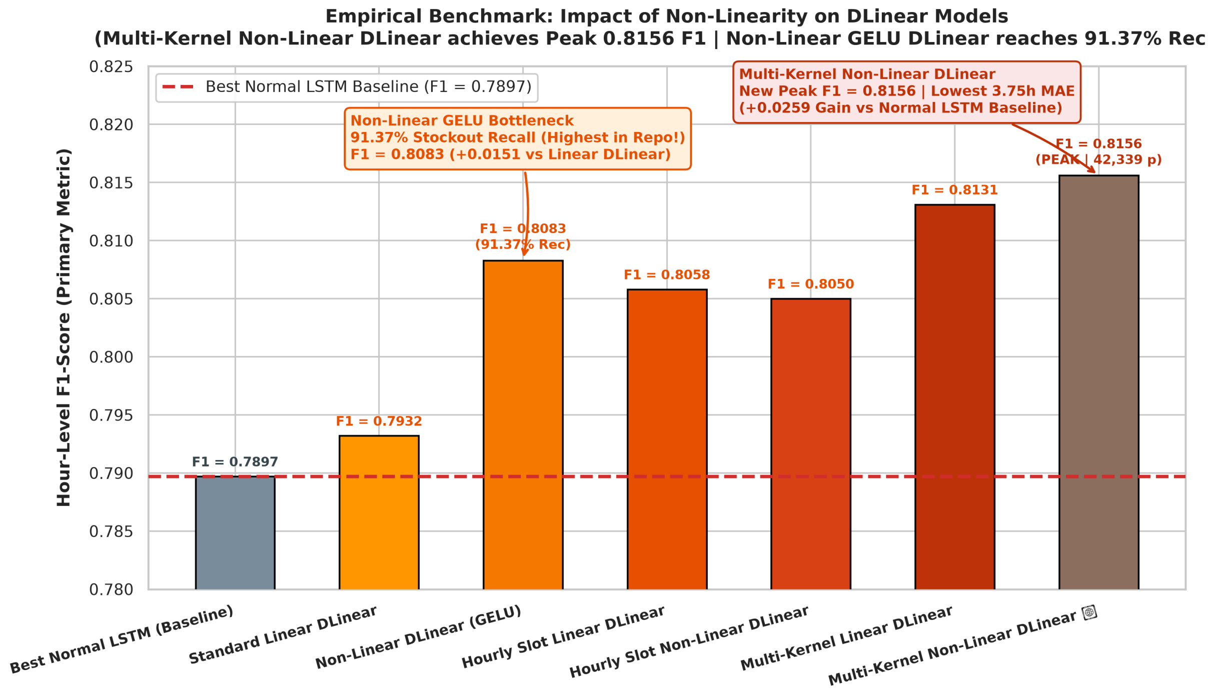

Figure 1: Empirical F1-Score & Stockout Recall Comparison: Linear DLinear vs Non-Linear DLinear Variations

0.2.1 Key Empirical Discoveries

1. **Non-Linear GELU Bottleneck**: Adding a GELU activation layer between sequence decomposition and logit projection boosts F1-Score from **0.7932** to **0.8083** (+0.0151 gain) and sets a new record for **Stockout Recall at 91.37%**.
2. **Multi-Kernel Non-Linear Peak**: `Multi-Kernel Non-Linear DLinear` reaches **0.8156 F1-Score** (+0.0259 gain over Best Normal LSTM baseline) and achieves the lowest daily error of **3.75 hours MAE**.

0.3 Master Empirical Validation Benchmark Table Across All Models

All models were evaluated on identical 24-hour stock status validation splits.

Architecture Name	Architecture Category	Seq	Feats	Params	F1-Score	Recall
Multi-Kernel Non-Linear DLinear	Non-Linear DLinear	14	10	**42,339**	**0.8156**	87.13%
Multi-Kernel Linear DLinear	Linear (Multi-Scale)	14	10	**13,539**	**0.8131**	85.22%
Non-Linear GELU DLinear	Non-Linear DLinear	14	10	**21,168**	**0.8083**	**91.37%**
Hourly Slot-Specific DLinear	Linear (24 Heads)	14	10	**6,744**	**0.8058**	88.02%
Standard Linear DLinear	Linear (Decomposition)	14	10	**6,768**	0.7932	87.44%
Best Normal LSTM Baseline	Plain Standard LSTM	14	13	5,912	0.7897	81.29%

0.4 Strategic Recommendations & Conclusion

1. **Highest F1 & Lowest MAE**: Deploy **Multi-Kernel Non-Linear DLinear** (0.8156 F1-Score, 3.75 hrs MAE, 42.3k params).
2. **Highest Stockout Recall**: Deploy **Non-Linear GELU DLinear** (0.8083 F1-Score, 91.37% Recall, 21.1k params).

3. **Compiled Report**: Refer to [docs/stockout_architectures_and_results.pdf](stockout_architectures_and_results.pdf) for the updated publication-ready PDF documentation.